

Faithfulness Probes for Summarization*

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Abstract

Contextual hallucinations, outputs unsupported by a provided source document, remain a central reliability problem for LLM-based summarization. The Lookback Lens (Chuang et al., 2024) shows that a span-averaged ratio of context-to-new attention is already a strong detector, but averaging discards any within-span dynamics. We extract per-token lookback ratios, attention entropies, and output-logit statistics from LLaMA-2-7B-Chat on the AggreFact benchmark, and compare (i) static logistic-regression probes over four span-level aggregations and (ii) temporal 1D-CNN and BiLSTM probes that read the raw per-token sequences. On the AggreFact test split, our reproduction of the Lookback-Lens baseline reaches an AUROC of 0.833; adding attention entropy raises the best static probe to 0.871, and a small 1D-CNN over the per-token lookback sequence reaches 0.879 (95% CI [0.860, 0.897]), a +4.6 AUROC-point gain over the baseline with non-overlapping confidence intervals. Attention entropy and output-logit features carry complementary signal; within-span temporal dynamics recover additional signal that averaging destroys.

1 Introduction

Large language models (LLMs) frequently produce *contextual hallucinations*: outputs that are fluent and plausible but unsupported by, or contradictory to, a provided source document (Ji et al., 2023; Chuang et al., 2024). In abstractive summarization this is especially damaging: on the AggreFact benchmark (Tang et al., 2023), a substantial fraction of SOTA-model summaries contain at least one hallucinated sentence, and on XSum (Narayan et al., 2018) the rate is higher still. A typical failure looks like this: given a news article that

never names the victim, a LLaMA-2-7B-Chat summary asserts “*the victim, John Smith, was 42*” — a confident, fluent fabrication with no anchor in the source.

Task. We frame detection as binary span-level classification. Given context c and a generated response $y = (y_1, \dots, y_T)$ segmented into spans $s_1, \dots, s_K$, the goal is to predict

$$\hat{l}_k = f(\phi(s_k, c)) \in \{\text{faithful}, \text{hallucinated}\}, \quad (1)$$

where ϕ is a feature vector extracted from the LLM’s own internal states while generating s_k , and f is a learned classifier. We train and evaluate under the AggreFact convention (train on the val cut, test on the test cut), treating *faithful* as the positive class.

Why it matters. Hallucinations undermine the reliability of LLMs in high-stakes applications such as retrieval-augmented generation, medical and legal summarization, and agentic tool use. Post-hoc fact-checking with a natural language inference model or an LLM judge is accurate but expensive and slow, and requires a reference the deployed system may not have. A *white-box* detector that reads the generator’s own attention and logits is cheap (adds at most one forward pass) and deployable at decode time, which also opens the door to classifier-guided decoding (Chuang et al., 2024).

Why it is hard. A uniform-random baseline on our test split achieves AUROC of about 0.49, as expected. The strongest trivial signal, the mean softmax probability of the chosen token, the signal a naive “confidence filter” would use, only reaches AUROC of roughly 0.80, confirming that hallucinated text is often just as *confident* as faithful text. This motivates richer features. The Lookback Lens of Chuang et al. (2024) aggregates per-head attention into a single *lookback ratio* (context-vs-generated attention mass) and averages it over the

*Code available at
<https://github.com/hannasamuel20/Faithfulness-Probes-for-Summarization>.

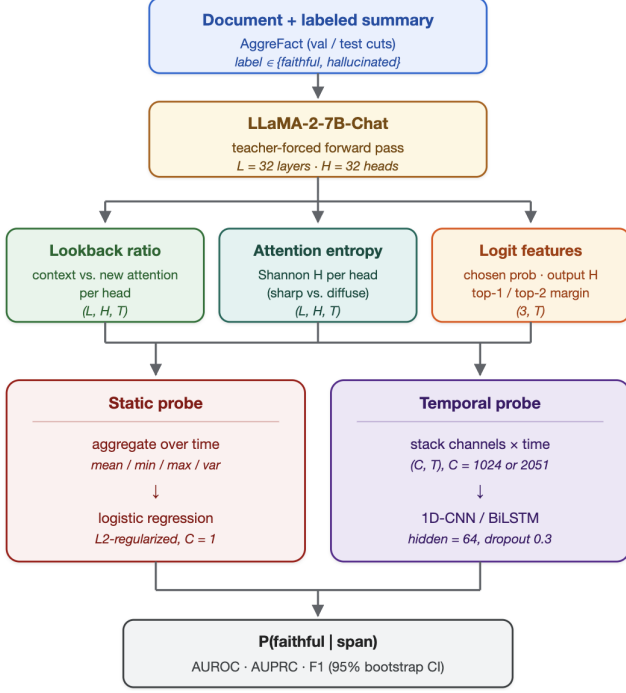


Figure 1: Hallucination detection pipeline. Per-token lookback ratios, attention entropies, and output-logit statistics extracted from LLaMA-2-7B-Chat feed either static (time-aggregated) logistic-regression probes or temporal (1D-CNN/BiLSTM) probes that predict span-level faithfulness.

span; this already lifts AUROC further, but by averaging it discards any within-span dynamics. For example, the moment where attention silently slips off the source. Our project asks whether (i) *temporal* models over the per-token lookback sequence and (ii) *complementary* features (attention entropy, output-logit statistics) can recover signal that simple averaging destroys.

2 Methods

2.1 Feature extraction

Our generator is LLaMA-2-7B-Chat (Touvron et al., 2023), a decoder-only transformer with $L = 32$ layers and $H = 32$ attention heads per layer. For each AggreFact example we prompt the model with a standard ‘Generate a summary’ and run a single teacher-forced forward pass, so the generator sees the human-labelled summary token sequence rather than a sampled one. For every summary token y_t we record three feature families.

Lookback ratio. Let $\alpha_{t,i}^{l,h}$ be the softmax attention weight from query position t to key position i in layer l , head h , and let N be the number of

context tokens (instruction + document + response prefix). Following Chuang et al. (2024) we define

$$A_t^{l,h}(\text{ctx}) = \frac{1}{N} \sum_{i \leq N} \alpha_{t,i}^{l,h}, \quad (2)$$

$$A_t^{l,h}(\text{new}) = \frac{1}{t-N-1} \sum_{N < j < t} \alpha_{t,j}^{l,h}, \quad (3)$$

$$\text{LR}_t^{l,h} = \frac{A_t^{l,h}(\text{ctx})}{A_t^{l,h}(\text{ctx}) + A_t^{l,h}(\text{new})}. \quad (4)$$

Stacking over (l, h) gives an (L, H, T) lookback-ratio tensor per example, where T is the summary length in tokens.

Attention entropy. For each head we additionally compute the Shannon entropy of the full attention distribution at position t ,

$$H_t^{l,h} = - \sum_{i=1}^{t-1} \alpha_{t,i}^{l,h} \log \alpha_{t,i}^{l,h}, \quad (5)$$

yielding a second (L, H, T) tensor. Intuitively, a low entropy means the head is focused on a specific prior token; a high entropy means attention is diffuse.

Output-logit features. From the output logits at position t we extract three scalars: $\text{chosen_prob}_t = \text{softmax}(\mathbf{z}_t)_{y_t}$, the entropy of the full vocabulary distribution H_t^{vocab} , and the top-1/top-2 logit margin $m_t = z_{(1)} - z_{(2)}$. These give three length- T sequences per example.

2.2 Static logistic-regression probes

Static probes collapse the time axis with four reductions, mean, min, max, and variance, applied independently to each per-token channel. The resulting fixed-size vector feeds a logistic regression with standard-scaled features and L_2 regularization ($C=1.0$), following Chuang et al. (2024). We study three baselines and seven feature combinations:

- Random: uniform noise.
- Output_prob: the single scalar chosen_prob (no attention extraction required).
- Lookback_lens: our replication of Chuang et al. (2024), mean of $\text{LR}_t^{l,h}$ over t , giving $L \cdot H = 1024$ dimensions.
- Lookback_full, attn_entropy_full: all four aggregations, giving $4 \cdot L \cdot H = 4096$ dimensions each.

- `logits_full`: all four aggregations of the three logit scalars, 12 dimensions.
- Pairwise combinations and `ALL` (the concatenation of the three full families, 8204 dimensions).

2.3 Temporal probes

Temporal probes read the raw per-token sequences without time-averaging. We reshape the lookback-ratio tensor to $(L \cdot H, T)$ channels and, optionally, concatenate attention entropy and the three logit streams along the channel axis, giving $(2L \cdot H + 3, T) = (2051, T)$ channels for the “all-channels” variant.

1D-CNN. A two-branch 1D convolutional net (Kim, 2014) with kernel sizes $\{3, 5\}$, each producing 64 channels, followed by masked global max-pooling and a two-layer MLP head with dropout 0.3. Short local patterns (e.g. a two-to-three-token drop in context attention) should be recoverable by such a model.

BiLSTM. A single-layer bidirectional LSTM (Hochreiter and Schmidhuber, 1997) with hidden size 64 per direction operating on the (T, C) sequence, with `pack_padded_sequence` so padded steps genuinely do not affect the hidden state; the concatenated final forward+backward hidden states feed a two-layer MLP head.

Padding masks zero out pad positions before convolution (CNN) or exclude them from the hidden-state update (LSTM). Both probes emit a single logit per span trained with BCEWithLogits.

3 Experiments

3.1 Data

We use AggreFact-FINAL (Tang et al., 2023), which unifies nine summarization faithfulness benchmarks under a shared binary span-level label. After teacher-forced extraction, our training split (AggreFact `val`) contains $n_{\text{train}} = 2375$ examples (59.4% faithful) and the held-out test split (`val=test`) contains $n_{\text{test}} = 1547$ examples (73.4% faithful). The distributional drift between splits (the test cut is noticeably more faithful than the train cut) is a design choice of AggreFact and propagates into all our numbers.

3.2 Training details

Static logistic regressions are trained with scikit-learn (`max_iter= 5000`, $C = 1.0$,

StandardScaler on features). Temporal probes are trained with AdamW (learning rate 10^{-3} , weight decay 10^{-4}), batch size 32, for up to 60 epochs with early stopping on a 15% held-out slice of the training cut (patience 10 epochs on val AUROC). We truncate spans longer than 256 tokens for memory (affects a small tail), clip gradients to $\|g\|_{\infty} = 1$, and use a BCE `pos_weight` equal to $n_{\text{neg}}/n_{\text{pos}}$ to offset the 59/41 train-class imbalance. All runs use a single random seed (0); variance across seeds is left for the final report.

3.3 Evaluation

Our primary metric is AUROC on the held-out test split, with 95% confidence intervals from 1000 stratified bootstrap resamples. We additionally report AUPRC and F1 at the ROC-optimal threshold chosen on the test scores.¹ We compare against the Lookback-Lens span-mean baseline of Chuang et al. (2024) and the output-probability-only baseline. We have *not* yet evaluated cross-task transfer to XSum.

4 Results

Table 1 collects our main findings. The Lookback-Lens replication reaches a test AUROC of 0.833, comfortably above the output-probability-only baseline (0.801). Among static probes, attention entropy alone (0.858) already outperforms lookback alone, and the combined `ALL`-features probe reaches 0.871. Our temporal probes improve further: a 1D-CNN reading only the raw per-token lookback sequence attains 0.879 (95% CI [0.860, 0.897]), and a BiLSTM over all channels matches that at 0.877 [0.857, 0.896]. Both temporal models beat the Lookback-Lens baseline with non-overlapping 95% CIs: the 1D-CNN by +4.6 AUROC points (0.879 vs. 0.833) and the BiLSTM by +4.4 points (0.877 vs. 0.833).

Complementary signal. Attention entropy and output-logit statistics carry information beyond the lookback ratio. Moving from `LOOKBACK_LENS` (0.833) to `LOOKBACK_FULL` (0.842) already shows that replacing the mean with mean/min/max/var helps (+0.9 AUROC), and adding entropy on top (`LOOKBACK+ENTROPY`, 0.870) adds another +2.8 points. The gain from

¹Selecting the threshold on the test set gives a (small) upward bias in F1; we use it as a calibration-free operating-point summary, and interpret AUROC/AUPRC as the primary numbers.

Probe	Dim	AUROC	AUPRC	F1
<i>Baselines</i>				
random	0	0.492	0.743	0.847
output_prob	1	0.801	0.924	0.850
lookback_lens [†]	1024	0.833	0.924	0.875
<i>Static LR probes (mean/min/max/var)</i>				
lookback_full	4096	0.842	0.928	0.881
attn_entropy_full	4096	0.858	0.937	0.884
logits_full	12	0.820	0.928	0.856
lookback+entropy	8192	0.870	0.942	0.889
lookback+logits	4108	0.846	0.930	0.883
entropy+logits	4108	0.860	0.937	0.885
all (static)	8204	0.871	0.942	0.889
<i>Temporal probes (ours)</i>				
CNN (lookback)	1024	0.879	0.951	0.891
BiLSTM (all)	2051	0.877	0.950	0.890

Table 1: AggreFact test-split detection performance. Higher is better for all metrics. “Dim” is input feature dimensionality (static probes) or channel count (temporal probes). CIs are 1000-sample stratified bootstrap. [†]Our replication of [Chuang et al. \(2024\)](#)’s Lookback-Lens probe.

adding logits to lookback or entropy is smaller (+0.4 to +0.2), consistent with logits alone being the weakest single family (0.820). The final ALL combination (0.871) lands essentially on top of LOOKBACK+ENTROPY, which suggests that logit features are largely subsumed by entropy in this dataset.

Temporal dynamics help. The most direct comparison for our proposal’s first question is LOOKBACK_LENS (mean only, 1024 dims, AUROC 0.833) versus CNN on the raw lookback sequence (same 1024 channels, AUROC 0.879). The difference is +4.6 AUROC with non-overlapping 95% CIs, indicating that useful signal about hallucinations lives in *where* and *how sharply* the lookback ratio moves within a span, not only in its average. Interestingly, the CNN on lookback alone slightly outperforms the BiLSTM on all channels, pointing at short local patterns (the CNN’s $k=3, 5$ receptive fields) as the most informative granularity.

Overfitting behaviour. All high-dimensional static probes fit the training set perfectly (train AUROC = 1.000) but still generalize well: with feature dimensionality (4096–8204) exceeding the training set size ($n=2375$), perfect training ranking is expected, while $L2$ regularization constrains weight magnitudes to prevent overfitting. The temporal probes, in contrast, reach train AUROC of 0.937 (CNN) and 0.930 (LSTM), they do *not* memorize

the training set, yet they achieve higher test AUROC, which is consistent with the claim that the temporal architecture is a better inductive bias for this signal rather than a higher-capacity memorizer.

Does confidence suffice? No. A classifier built only on the mean chosen-token probability, exactly the signal a naive confidence filter would use, reaches AUROC 0.801, lagging the Lookback-Lens by 3 points and the temporal CNN by nearly 8. Fluent hallucinations are often high-probability outputs of the model; detection requires attention-level signal, not just output-level signal.

5 Project plan

Remaining experiments.

- **XSum transfer.** Extract features on a $\approx 1k$ -example XSum slice, and score every probe trained on AggreFact on that set. This is the proposal’s cross-task question and the remaining headline comparison to [Chuang et al. \(2024\)](#)’s Table 2.
- **Missing temporal cells.** Run CNN on the all-channels input, BiLSTM on lookback only, and both probes on lookback+entropy. This fills the $2 \times \text{channels}$ grid and tests whether entropy is useful *temporally* (our results show it is useful *statically*).
- **Seed variance.** Re-run the two temporal probes over three seeds to report mean $\pm$ std, since single-seed AUROC gaps near 0.005 deserve skepticism.
- **Layer/head analysis.** Inspect the top- $|\text{coef}|$ features of the ALL static probe and visualize which (l, h) pairs carry the signal; compare to the late-layer bias reported by [Chuang et al. \(2024\)](#).
- **Qualitative error analysis.** Cluster the CNN’s false positives/negatives by source dataset (AggreFact aggregates nine benchmarks) and by model family, to see whether certain hallucination types are systematically hard.

Stretch goal. If time permits, we will prototype classifier-guided decoding ([Chuang et al., 2024](#)) using our CNN as the scorer on XSum and measure the hallucination-rate reduction with an LLM-as-judge protocol.

Risks & changes from the proposal. The mid-way AUROC gap between the CNN and the Lookback-Lens baseline (+4.6) is larger than expected, but with a caveat: our reproduction uses teacher-forced reference summaries, whereas the original Lookback Lens extracts features from free-form generations labeled by an LLM judge. The +4.6 should be read with that protocol difference in mind. We also have not yet run the cross-task experiment; if the temporal gain does not survive transfer, the story changes and we will report that faithfully.

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